

Filtering clinical-guideline sections before the LLM reads them

A two-stage condition-matching approach for Russian medical text.

55.17 / 70 M1 0.922 · M2 0.731

`solution.ipynb`: reproducible in Colab, ~1.5 min on CPU, no GPU

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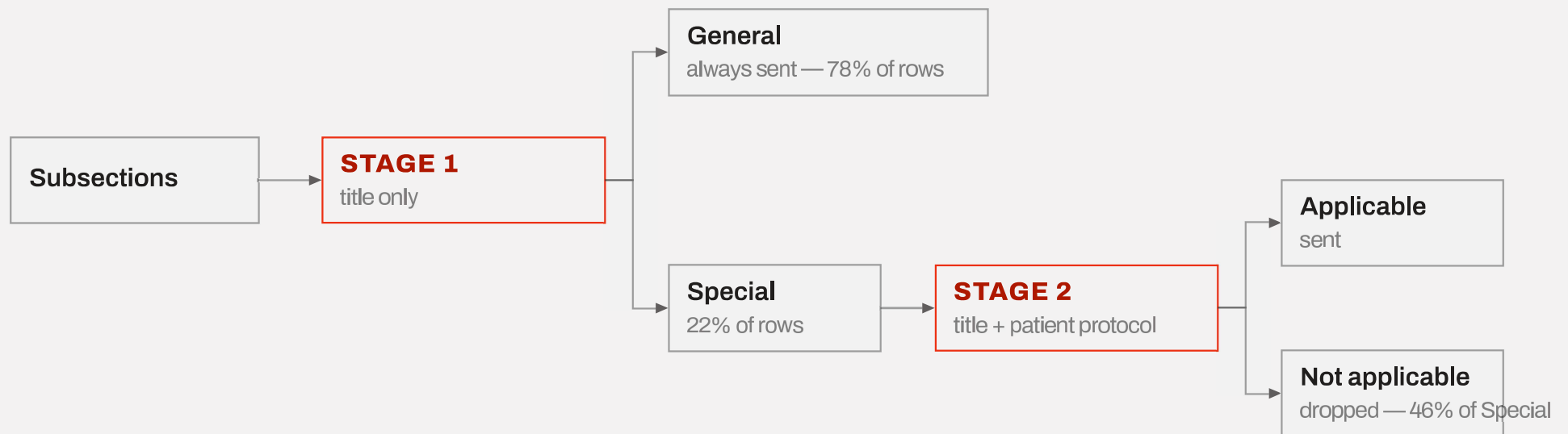
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01

The task

Every Treatment subsection we can safely drop saves LLM tokens. Every subsection we drop by mistake can hide a relevant recommendation.

The two filters



Stage 1 asks whether a subsection applies to everyone or to a restricted group; Stage 2 asks whether a restricted group includes this patient. Roughly 10% of subsections per patient are removed by count.

The metric and the cost of each error

ERROR	CONSEQUENCE	HOW THE METRIC TREATS IT
Stage 2 drops an applicable section	The LLM never sees a relevant recommendation — a clinical risk	F2: recall weighted 4×
Stage 2 keeps an inapplicable section	Wasted tokens	tolerated
Stage 1 calls a General section Special	The section can then be dropped in Stage 2	F0.5: precision weighted 4×

Score = $70 \cdot (0.3 \cdot M1 + 0.7 \cdot M2)$. One extra hundredth of M2 is worth 2.3× one extra hundredth of M1, so about 70% of the work went into Stage 2. The asymmetry also explains both threshold decisions later in the deck.

The data

	STAGE 1	STAGE 2
Train / test rows	1,767 / 442	690 / 173
Class balance	78% General / 22% Special	66% Applicable / 34% Not applicable
Unique units in train	491 guidelines	37 titles, 221 protocols
Median text length	title only	≈4,800 characters per protocol

690 pairs over 37 titles is the binding constraint: it rules out fine-tuning, and it sets our noise floor.

Three properties that shaped the design

Small data **± 0.02**

M2 moves by this much from the choice of a single 5-fold split alone. 690 pairs are too few to fine-tune a transformer reliably.

Lab tables dominate **78%**

of all protocol characters are lab lines (Показатель ... Результат ...). 187 of 863 texts are cut at exactly 32,765 characters.

Title-heavy labels **0.874**

in-sample AUC from the per-title Applicable rate alone. 13 of 37 titles are all-Applicable or all-Not-applicable.

But the label is not the title: 102 protocols appear under more than one title, and 62 of them get different labels — so the decision lives in the (title, patient) pair.

The damaged training file

- The first public copy of **train_stage1.csv** had every Cyrillic letter replaced by ? — about 84% of the characters were lost.
- The first three Stage 1 generations were built on that copy, using corruption-invariant features: word shapes, then partial Cyrillic decoding (M1 0.711, then 0.753).
- A clean re-download let the model read real words and lifted Stage 1 from **M1 0.753 to 0.843** in nested CV.
- The notebook now **refuses to run** on the damaged file rather than degrading silently.

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The two models

Two transparent linear models. Each one encodes the task definition directly as features, rather than hoping a larger model will infer it.

The solution at a glance

Stage 1 — condition detection

Parse the heading hierarchy, then four feature blocks: lemmatised last line, guideline and parent context, char 3–5-grams, layout and condition markers.

Logistic regression, $C = 10$, balanced, threshold 0.69, plus the Stage-2-title rule.

Stage 2 — condition matching

From the title: age bound, sex, severity level, disease terms. From the protocol: Пол, Возраст, the narrative sections, negations.

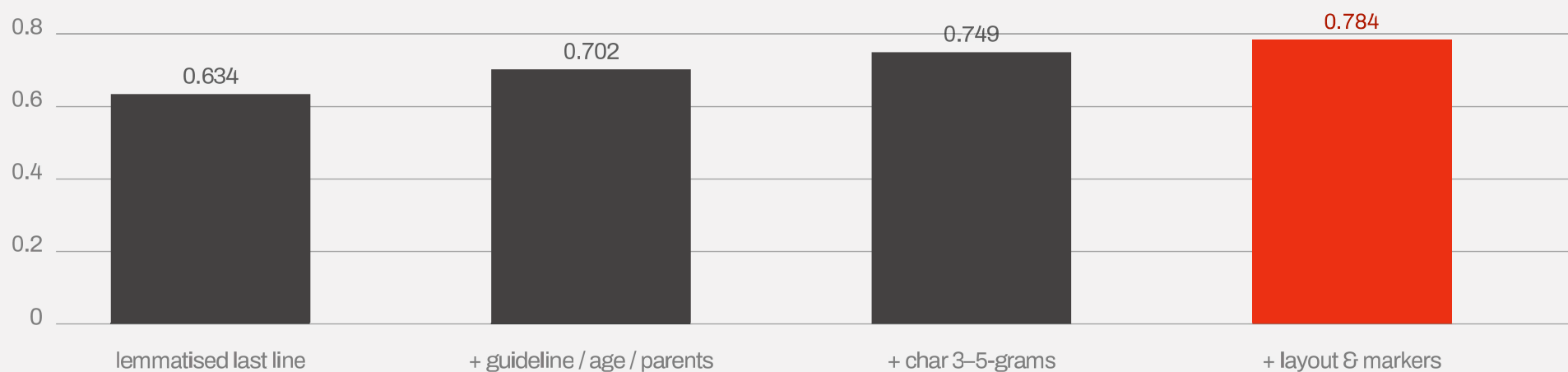
Each condition resolves to satisfied, contradicted or unknown — 11 features joined with title and narrative TF-IDF. Logistic regression, balanced, threshold 0.5.

Stage 1: M1 0.843 in nested CV → 0.922 on the leaderboard. Stage 2: M2 0.702 → 0.731. Points 52.1 → 55.17.

Stage 1: detecting a restricted population

- “Special” means **the title names a restricted population**, an age bound, a sex or pregnancy, a disease form, localisation or stage, a comorbidity, or a treatment line such as relapsed or refractory disease.
- Condition-marker lemma sets cover seven groups: age, sex and pregnancy, severity, form, comorbidity, refractory disease, restriction words.
- The hierarchy itself is evidence: the Special rate is **15% at heading depth 2** and **52% at depth 3**.
- Marker words point towards Special but never settle it, which is why they are one block among four and not a hard rule.

Stage 1: contribution of each feature block



Stage 1: threshold and the Stage-2-title rule

Threshold 0.69, chosen on inner folds

Nested CV picks C and the threshold: the inner folds choose, the outer folds score. The five inner choices were 0.73, 0.64, 0.69, 0.65 and 0.79; the submitted value is their median.

The Stage-2-title rule

Stage 2 contains only Special subsections, so any title that also occurs in the Stage 2 data must be Special. On train: 32 matches, 100% precise.

On test it matched 5 titles and changed 0 predictions. It uses titles only, never labels: a safety net, not a score source.

Nested out-of-fold Stage 1 quality: precision on Special 0.84, recall 0.75: 95 false negatives and 55 false positives out of 1,767 rows.

“A section is Not applicable only when the patient contradicts the title’s condition. Missing evidence keeps it applicable.”

.....the Stage 2 design principle: unknown $\neq$ contradiction

Stage 2: what the condition states predict

STATE OF THE TITLE’S CONDITION AGAINST THE PATIENT	ROWS	SHARE APPLICABLE
Age contradicted (до N лет vs the patient’s age)	17	0.00
Sex-specific title, sex known from Пол:	51	0.04
Sex-specific title, sex unknown or pregnancy title	18	0.33
Disease term never mentioned / unknown	214	0.55
Disease term present, not negated	418	0.71
Disease term present and negated	58	0.71
All rows	690	0.66

Unknown sex keeps 33% of rows applicable against 4% when sex is known; the negation flag carries no signal.

Stage 2: title conditions against patient facts

TITLE CONDITION	PATIENT FACT	FEATURES
Upper age bound до N лет	Age — regex with a free-text fallback	age_contradiction
Female / male / pregnancy topic	Structured Пол field only	gender_condition_present, gender_unknown
Severity level (mild / moderate / severe)	Severity words in the narrative	severity_match, _contradiction, _unknown
Informative disease terms	Narrative lemmas plus a negation window	condition_overlap, _contradiction, _unknown
Aggregate over all conditions	—	contradiction_count, contradiction_density

Every condition carries separate contradicted and unknown features, so missing information can never look like a contradiction to the model. Sex is read only from the structured field: guessing it from пациентка created false contradictions.

Russian medical text handling

PROBLEM	WHAT WE DO	WHERE
Rich morphology (тяжёлой / тяжелая)	pymorphy3 lemmatisation, cached; char n-grams catch the rest	both
Meaningful function words (при, у пациентов с)	Stage 1 keeps every word; Stage 2 removes stop words	both
Patient-specificity vocabulary	Lemma sets for seven condition groups	Stage 1
Negation (не, нет, отсутств..., отрицательн...)	Count per protocol plus a 40-character window per term	Stage 2
Structured fields in free text	Regex for Пол, Возраст, Жалобы / Анамнез / Объективный статус	Stage 2
Severity terms	Mapped to levels, compared between title and narrative	Stage 2
Long protocols	Field extraction, not truncation — the narrative is 19% of the text	Stage 2

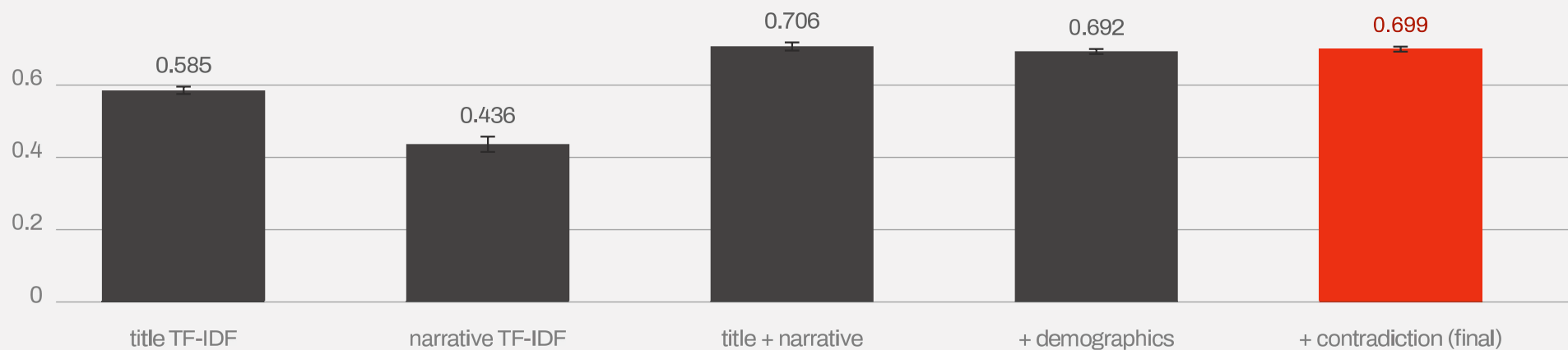
In «Жалоб нет. Отёков не выявлено, беременность отрицает.» only не выявлено matches. The final нет. and отрицает are missed.

03

Evidence

Stage 2: contribution of each component

M2, 5 × 5 repeated CV · mean ± sd



Why the contradiction model is still the submission

- The leaderboard verified it at **55.17**, and no alternative beats it under 5×5 repeated CV.
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- Switching on a 0.006 difference would be selection on noise, which is the mistake the harness exists to prevent.
- What we learned about our own process: a single 5-fold split cannot detect gains below about 0.02 M2. We built a 5×5 repeated-CV harness with a corrected paired t-test and a noise-feature control, and used it for every later decision.

Rejected hypotheses

#	HYPOTHESIS	RESULT	VERDICT AND REASON
T2	rubert-tiny2 on title + narrative	M2 0.496	X 690 rows are too few to fine-tune
T4	Structured-field template for the transformer	M2 0.540	X best transformer, still -0.14
H7	Raise the Stage 2 threshold to 0.54	M2 0.706	X buys M2 by giving up Applicable recall
H8	Explicit contradiction / unknown features	M2 0.702	✓ promoted — +0.023 on the fixed split
H22	Threshold tuned inside nested CV	$\Delta -0.014$	X 0.5 is already the right cut
H24-B/C	+ ICD-10 code and diagnosis text (C0)	$\Delta +0.022$, $p = 0.14$	X leaderboard $54.80 < 55.17$
H24-D	Zero-shot Qwen2.5-7B probability as a feature	AUC 0.735 alone	X positive-biased; within-title AUC 0.52

Also rejected: head + tail truncation (0.477), finer negation scope (-0.036), recurrence counts, per-title calibration, CatBoost.

Three takeaways from the failures

- **Model capacity is not the bottleneck.** Transformers, CatBoost and a 7B LLM all lost under identical splits. The bottleneck is labelled evidence per title.
- **Rules help only when they are precise.** Age and sex rules work because they read structured fields; free-text negation windows do not.
- **The CV-to-leaderboard check matters.** The ICD model looked like the best CV result of the project and still scored 0.37 points lower; we kept the verified model.

04

Limits and next steps

55.17 is the score on titles the model has already seen. Deployment would be harder, and the remaining errors say why.

Remaining errors, by kind

Stage 1 · missed Special — 95

57 have no condition marker at all: the population is named implicitly, through a situation or a disease variant (повторное хирургическое вмешательство).

Stage 1 · false Special — 55

33 do contain a marker, used generically in a heading the annotators read as general (лечение осложнений). Only 4% of General titles are exposed.

Stage 2 · false negatives — 81

73% happen because the title's low prior dominates the score; 50 of the 81 sit in seven titles. Recall on Applicable is 0.82.

Stage 2 · false positives — 42

Precision on Applicable stays 0.90 — these cost tokens, not recommendations, which is the cheaper of the two errors.

Where the Stage 2 errors concentrate

TITLE (LAST LINE)	ROWS	SHARE APPLICABLE	FN	WITHIN-TITLE AUC
Tension headache with medication-overuse headache (ЛИГБ)	21	0.48	10	0.22
Surgery for Crohn's disease, terminal ileitis	19	0.74	9	0.71
Perianal Crohn's disease	17	0.41	7	0.31
Erosive gastritis / duodenitis	20	0.80	7	0.34
Crohn's disease, upper GI tract	22	0.32	7	0.74

Seven titles hold 50 of the 81 false negatives; within three of them the model ranks below chance. 46% of all false negatives are pulled down by the title's low prior, 12% are negation-window false alarms.

A concrete case: perianal Crohn's disease

Patient A — dropped, wrongly

«Болезнь Крона, язвенный колит с 14–15 лет. Анемия.» (K50.8).

Perianal disease is **not mentioned**.

True label Applicable, nothing contradicts БК с перианальными проявлениями. The model scored 0.14.

Patient B — kept, wrongly

«Болезнь Крона около 4 лет...» (K50.9), a normal examination.

True label Not applicable. The model scored 0.73.

- The decisive fact is often an **absence**, and a bag-of-words model scores what is present.
- Within such a title the model has only 17–22 examples, the limitation is structural, not a matter of tuning.

0.42

M2 when the title is held out of training, against
0.702 when it is not.

All 32 test titles and 89% of test protocols also appear in the training data. The model works mostly as a per-title prior refined by the patient's text, so it does not yet transfer to a new guideline.

The competition metric is not clinical safety

THRESHOLD	M2	APPLICABLE RECALL	POSITIVE RATE
0.30	0.539	0.934	0.77
0.40	0.622	0.876	0.69
0.50 -> submitted	0.702	0.821	0.60

At the metric-optimal point, 18% of applicable special sections are still dropped. For deployment we would set a recall target of about 0.93, accept lower token savings, and log every dropped section. The saving is modest and unmeasured in any case: roughly 10% of subsections per patient by count, and the data carries no section lengths to convert that into tokens.

Next steps, in priority order

- | | | |
|----|--|---|
| 01 | Condition extraction as NER with assertion status | replaces the negation window; represents absence explicitly |
| 02 | Title parsed into structured criteria | moves decisions from title identity to criteria vs facts |
| 03 | An LLM as a verifier, not a feature | called only on low-confidence rows, calibrated per title |
| 04 | More labels for low-prevalence titles | active learning -> the most reliable lever |
| 05 | A safety operating point, and measured token savings | what the competition did not reward |
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Reproducibility

- **One notebook, 47 cells**, from the four CSVs alone: setup, data, metric, EDA, text processing, both stages, submission. About 1.5 minutes on a CPU, no GPU.
- **Pinned and seeded**: scikit-learn 1.9.0, pymorphy3 2.0.6 with its dictionary, seed 42 everywhere.
- **The notebook checks itself**: it re-derives C and the threshold, re-checks the Stage 2 score of 0.7024, and compares a SHA-256 fingerprint of each stage's predictions — stopping if they differ.
- **Fold safety**: vectorisers and scalers fit inside the pipeline per fold, contradiction features are row-local, and a leak in our own recurrence experiment was caught and re-run.
- **Every rejected idea is kept**: one folder per hypothesis with its script, results table and report.

Condition detection, then condition matching

55.17 / 70 from two transparent linear models that beat transformers, CatBoost and a 7B LLM in controlled tests with 18% of applicable sections still dropped, no transfer to unseen titles yet, and a negation feature that carries no signal.

Next: assertion-aware NER · structured title criteria · an LLM verifier · more labels

Stage 1 model history

MODEL	TRAINING TEXT	VALIDATION M1	LEADERBOARD M1
char TF-IDF + 6 meta features → LinearSVC	damaged copy	0.465	0.629
rubert-tiny2 fine-tune (T1)	damaged copy	0.458	—
Word-shape corruption-invariant features (H14)	damaged copy	0.711	—
Invariant features + partial Cyrillic decoding (H15)	damaged copy	0.753	—
Lemmatised word + char + condition markers (H17)	clean	0.843	0.922

Validation figures are nested CV from H14 onwards.

Submissions and score decomposition

The three submissions

Stage 1 v1 + G — 49.02.

H17 + G, the final model — 55.17.

H17 + C0, adding the ICD code and diagnosis text, 54.80. It changed only 6 of 173 test predictions, all towards “applicable”, and still cost 0.37 points.

Where the headroom is

Stage 1 contributes 19.36 of 21 possible points — 1.64 points of headroom.

Stage 2 contributes 35.81 of 49 — 13.19 points of headroom.

No hyperparameter, threshold or feature was tuned on leaderboard feedback; the only leaderboard-informed decision was choosing between two pre-built models.

The validation protocol

- 5×5 repeated stratified K-fold with paired fold-level differences, and a Nadeau–Bengio corrected t-test for overlapping training sets.
- **Gate:** mean $\Delta > 0$, one-sided $p < 0.10$, and at least 4 of 5 repeats improved.
- **Negative control:** a pure-noise feature scored $\Delta -0.012$ with 0 of 5 repeats improved, and the gate rejected it.
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- Measured noise: one 5-fold split varies by ± 0.02 M2, the 5×5 standard deviation is about 0.007, and the Stage 1 nested folds range from M1 0.79 to 0.90.

Where the numbers come from

NUMBERS	SOURCE
Final score, leaderboard M1 / M2, points decomposition	h18/metric_decomposition.json
C0 leaderboard 54.80; harness, CatBoost, LLM feature	h24/H24_REPORT.md, h24/step_*_results.csv
EDA — lengths, lab share, title AUC, mixed labels	solution.ipynb §3
Stage 1 ablation, nested CV, errors	solution.ipynb §5.3–5.7
Stage 2 state table, ablation, threshold sweep, per-title errors	solution.ipynb §6.2–6.7
False-negative categories	h20/fn_categories.csv
Transformer runs T1–T4, H1	H6_control_summary.json, baseline.ipynb
H19–H22, including grouped-by-title 0.422; Stage 1 history	h22/FINAL_REPORT.md; h14/, h15/, h17/ manifests